☆ $Current\ (I) = \frac{Q}{t} = \frac{\text{Coulomb}}{\text{seconds}}$ [Amps]

☆ $Voltage\ (V) = \frac{W}{Q} = \frac{\text{Joules}}{\text{Coulomb}}$ [Volts]

☆ $Power\ (P) = \frac{W}{t} = \frac{\text{Joule}}{\text{Sec}}$ [Watt / Calories]

Resistance from Watts: $R = \frac{V^2}{P}$

Power (P) = 1A * 1V [Watts]

Energy (Watt hour) = Power * time [Wh]

Energy (kWh) = $\frac{\text{Power[watt]*time[hour]}}{1000}$

Resistance of a Wire: $R[\text{Ohms}] = \frac{\rho[\text{ohm-meters}]L[\text{meters}]}{A[\text{meters}^2]}$

Where $A = \pi\left(\frac{d}{2}\right)^2$ if d in meters, or $A_{CM} = (d_{mils})^2$

Resistivity of Copper Wire at 20C : $1.68 * 10^{-8}\Omega\text{m}$

Resistance of a Conductor at Varying Temperature:

$R = R_{\text{ref}}[1 + \alpha(T - T_{\text{ref}})]$ where α is located in T3.6

Efficiency: $W_{\text{in}} = W_{\text{out}} * W_{\text{lost}}$ & $P_{in} = P_{out} + P_{lost}$

W= P*t = V*Q → I=Q/T → Q=I*t →W=V*I*t

Resistance from T, use Celsius: $\frac{|T_i|+T_1}{R_1} = \frac{|T_i|+T_2}{R_2}$, use T3.5

Kirchoff's Voltage Law (KVL): $\Sigma V = 0$

Reminder: Circuits with a single voltage source, current exits from (+) of power source and enters the (-) side.

Reminder: Voltage rises going through (-)(+) pair and decreases when going through a (+)(-) pair.

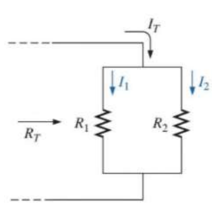
Kirchoff's Current Law (KCL): $\Sigma I_i = \Sigma I_0$

Definition: Current proportionally divides and flows like a river stream.

Voltage Divider (To find V through a resistor)

$$V_{out} = V_{in}\left[\frac{R_2}{R_1+R_2}\right] \qquad V_1 = \frac{R_1 E}{R_T} = \frac{R_1 E}{R_1+R_2+R_3}$$

Current Divider (To find I through a resistor)



$$I_x = \frac{R_T}{R_x} I_T \qquad I_x = \frac{R_1}{R_1+R_2} I_T$$

Equivalent Resistors

Fact 1: A resistor that isn't connected in a closed loop can be neglected.

Fact 2: Current flows from (+) to (-) in resistors.

Fact 3: 2 grounds mean they're connected.

Sanity Checks:

☆ The equivalent resistor is smaller than the smallest resistor by ~20%.

☆ If one of the resistors is **significantly less** than all the other resistors, then R_{total} will be very close to that smallest resistor value.

Series Circuits

Definition: R1 and R2 connected, resulting in one and only one connection with each other.

☆ **Fact 4:** All currents are the same through all elements in series.

Fact 5: The total resistance is unaffected by the order in which resistors in series are connected.

Fact 6: In series, batteries can be rearranged.

Current in Series: $I_s = I_1 = I_2 = I_3$

Total Current in Series: $I_s = \dfrac{V}{R_1 + R_2 + R_3} = \dfrac{V}{R_T}$

Total Resistance in Series: $R_T = R_1 + R_2 + R_3$

Total Resistance in Series: $R_T = \dfrac{E}{I}$

Voltage Sources in Series: $E_T = E_1 + E_2 + E_3$

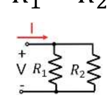
Parallel Circuits

Definition: Two elements, branches, or circuits are in parallel if they have two points in common.

☆ **Fact 7:** All voltages are the same through all elements in parallel.

Voltage in Parallel: $V_s = V_1 = V_2 = V_3$

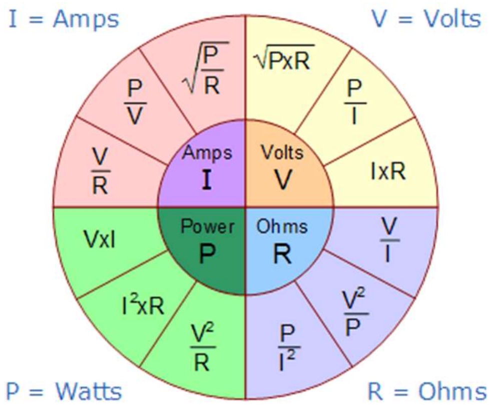
Total Resistance in Parallel: $R_T = \dfrac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}}$

Special Case: $R_T = \frac{R_1 R_2}{R_1+R_2}$ when 

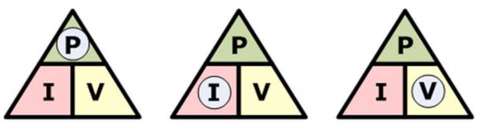
Conductance: $G = G_1 + G_2 + \cdots + G_N, where\ G = 1/R$

Total Current in Parallel: $I_S = \dfrac{V}{\dfrac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}}}$

Ohm's Law Pie Chart:

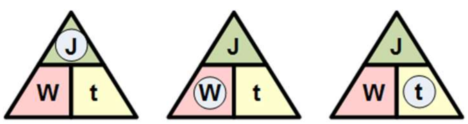


Power, Current, Volts:



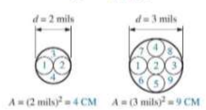
P = I x V   I = $\frac{P}{V}$ V = $\frac{P}{I}$

Joules, P (Watts), time:



J = W x t   W = $\frac{J}{t}$ t = $\frac{J}{W}$

- One mil is a thousandth of an inch:
  1000 mil = 1 in
- A circle with a diameter of one mil has an area of one Circular Mil (CM
  $A_{CM} = (d_{mils})^2$



$$A = \frac{\pi d^2}{4} = \frac{\pi}{4}(1\ mil)^2 = \frac{\pi}{4}\ sq\ mils = 1\ CM$$

TABLE 2.1
Relative conductivity of various materials.

Metal	Relative Conductivity (%)
Silver	105
Copper	**100**
Gold	70.5
Aluminum	61
Tungsten	31.2
Nickel	22.1
Iron	14
Constantan	3.52
Nichrome	1.73
Calorite	1.44

TABLE 3.6
Temperature coefficient of resistance for various conductors at 20°C.

Material	Temperature Coefficient (α_{20})
Silver	0.0038
Copper	0.00393
Gold	0.0034
Aluminum	0.00391
Tungsten	0.005
Nickel	0.006
Iron	0.0055
Constantan	0.000008
Nichrome	0.00044

TABLE 3.2
American Wire Gage (AWG) sizes.

	AWG #	Area (CM)	Ω/1000 ft at 20°C	Maximum Allowable Current for RHW Insulation (A)*
(4/0)	**0000**	211,600	0.0490	**230**
(3/0)	**000**	167,810	0.0618	**200**
(2/0)	**00**	133,080	0.0780	**175**
(1/0)	**0**	105,530	0.0983	**150**
	1	83,694	0.1240	**130**
	2	66,373	0.1563	**115**
	3	52,634	0.1970	**100**
	4	41,742	0.2485	**85**
	5	33,102	0.3133	—
	6	26,250	0.3951	**65**
	7	20,816	0.4982	—
	8	16,509	0.6282	**50**
	9	13,094	0.7921	—
	10	10,381	0.9989	**30**
	11	8,234.0	1.260	—
	12	6,529.9	1.588	**20**
	13	5,178.4	2.003	—
	14	4,106.8	2.525	**15**
	15	3,256.7	3.184	
	16	2,582.9	4.016	
	17	2,048.2	5.064	
	18	1,624.3	6.385	
	19	1,288.1	8.051	
	20	1,021.5	10.15	
	21	810.10	12.80	
	22	642.40	16.14	
	23	509.45	20.36	
	24	404.01	25.67	
	25	320.40	32.37	
	26	254.10	40.81	
	27	201.50	51.47	
	28	159.79	64.90	
	29	126.72	81.83	
	30	100.50	103.2	
	31	79.70	130.1	
	32	63.21	164.1	
	33	50.13	206.9	
	34	39.75	260.9	
	35	31.52	329.0	
	36	25.00	414.8	
	37	19.83	523.1	
	38	15.72	659.6	
	39	12.47	831.8	
	40	9.89	1049.0	

TABLE 3.1
Resistivity (ρ) of various materials.

Material	ρ (CM · Ω/ft)@20°C
Silver	9.9
Copper	**10.37**
Gold	14.7
Aluminum	17.0
Tungsten	33.0
Nickel	47.0
Iron	74.0
Constantan	295.0
Nichrome	600.0
Calorite	720.0
Carbon	21,000.0

TABLE 3.5
Inferred absolute temperatures (T_i).

Material	°C
Silver	−243
Copper	−234.5
Gold	−274
Aluminum	−236
Tungsten	−204
Nickel	−147
Iron	−162
Nichrome	−2,250
Constantan	−125,000

TABLE 3.3
Resistivity (ρ) of various materials.

Material	Ω-cm
Silver	1.645×10^{-6}
Copper	1.723×10^{-6}
Gold	2.443×10^{-6}
Aluminum	2.825×10^{-6}
Tungsten	5.485×10^{-6}
Nickel	7.811×10^{-6}
Iron	12.299×10^{-6}
Tantalum	15.54×10^{-6}
Nichrome	99.72×10^{-6}
Tin oxide	250×10^{-6}
Carbon	3500×10^{-6}

Generalized steps to solve circuit:

1. Determine R_T
2. Calculate R_S
3. Determine R_X (resistors' R)
4. Find E
5. Calculate P

Solving series from sample exam:

1. Refer to Fact 4.
2. Move E sources together and add them.
3. Combine resistors.
4. Current direction and polarities
5. Use voltage divider for Rx

Solving parallel from sample exam:

1. Find RT using Equation.
2. Use Fact 7 to find I.
3. Use current divider to find Rx.

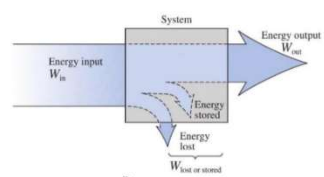
Solving circuits checklist:

Do you have your…

- Voltages?
- Resistor Polarity?
- Current direction?
- Resistances?
- Currents?
- Power (not sure if req'd?)

Solving cascaded efficiency:

$$\eta_T = \eta_1 * \eta_2 * \ldots * \eta_n$$



$$\eta = \frac{P_{Out}}{P_{in}}$$

$$\eta\% = \frac{P_{Out}}{P_{in}} \times 100\%$$

Other things:

☆ Ohmmeters and Ammeters read (+)(-), negative values if switched.

☆ Use KVL on the path if solving voltage between 2 arbitrary points

☆ Remember to block approach!